

The Crisis of 1900

Three facts, each of them solid. Any two of them are comfortable together. All three cannot be true at once, and this chapter is the proof.

WHERE WE ARE

Part I rebuilt mechanics from an action principle and it worked. A single scalar function, varied over paths, gave back Newton's second law, gave it back in *any* coordinates, handled constraints without ever naming a constraint force, and — through Noether — turned every symmetry into a conservation law. The machinery is sound. Nothing in Part II takes any of it back.

Now we point it at light, and it breaks.

Here is the shape of the chapter, stated in advance so you can watch it close. There are three claims.

(1) The laws of physics take the same form in every inertial frame, and the transformation relating those frames is $x' = x - vt$, $t' = t$ — this is Galilean relativity, it is older than Newton, and Newton's laws satisfy it exactly. **(2)** Maxwell's equations, which describe every electric and magnetic phenomenon anyone had measured by 1890, imply that electromagnetic disturbances propagate at a definite speed c built out of two laboratory constants. **(3)** Experiment finds no trace of the frame that (1) and (2) together demand must exist.

Each claim is going to be established here, not asserted: (1) by direct calculation in §1, (2) by deriving the wave equation from Maxwell in §2, (3) by working out what Michelson and Morley should have seen in §5 and comparing it to what they did. Section 4 is the hinge — it shows by explicit chain rule that Maxwell's equations *change form* under the transformation in (1), which is where the contradiction actually lives.

One thread is picked up from earlier. Chapter 1.1 §3.3 showed that Newton's third law fails for two charges in relative motion: an isolated pair of particles has $d\mathbf{P}_{\text{mech}}/dt \neq \mathbf{0}$, at a computable rate suppressed by v^2/c^2 . The only available repair was that the field carries the missing momentum, and the suspicious factor v^2/c^2 was left hanging. That same factor is about to reappear as the size of the effect Michelson and Morley went looking for. It is the same v^2/c^2 , and it is not a coincidence: both are the leading signature of the fact that electromagnetism does not transform the way mechanics does.

What this chapter does not do. It does not derive the Lorentz transformation. That is Chapter 2.2's job, and doing it early would spoil the only honest reason to accept it — which is that by the end of §7 you will want *some* transformation that resolves the contradiction, and there turns out to be essentially one candidate. The chapter ends with two postulates and nothing else.

TOOLS YOU'LL NEED — Chapter 0.3: the binomial series, and what "to leading order in β^2 " means quantitatively. Chapter 0.6 §5.2: the multivariable chain rule for a change of coordinates — §4 is nothing but that, ground out. Chapter 0.7 §4 (curl), §3 (divergence), §7 (the Laplacian and the second-derivative identities): all four Maxwell equations are written in that language and §2 manipulates them there. Chapter 0.8 §7.6: the wave equation, its solutions $f(x - vt)$, and the fact that a wave equation's speed is a property of the medium. Chapter 1.1 §3.3: the third law failing for moving charges, and the loose end that failure left.

1 · Galilean relativity, stated precisely

The idea is four hundred years old and it is correct. It deserves to be stated properly rather than waved at, because the whole crisis turns on the difference between the *principle* and the *transformation that implements it* — two things that were fused together for three centuries and that Chapter 2.2 will finally prise apart.

1.1 · The transformation

Two observers. Call their frames S and S' . They agree on how to measure lengths and they each carry a clock. At $t = 0$ their origins coincide, and thereafter S' moves in the $+x$ direction with constant speed v as measured in S . This arrangement — axes aligned, relative motion along the shared x -axis, origins coincident at zero — is called **standard configuration** and it is the setting for everything in Part II.

Ask where an event is, in each frame. The event happens at position x at time t according to S . Since S' 's origin has by then slid a distance vt along the axis, the same event is a distance $x - vt$ from that origin. And — this is the assumption, put in explicitly so we can watch it die — both observers assign the event the same time. Hence the **Galilean transformation**:

$$x' = x - vt, \quad y' = y, \quad z' = z, \quad t' = t. \quad (2.1.1)$$

Read the last equation carefully. $t' = t$ is not a triviality and it is not bookkeeping. It is a substantive physical claim: *there is a single universal time, ticking at the same rate for everyone, and two events either are simultaneous or are not, with no reference to who is asking*. Newton stated it in exactly those terms — absolute time, flowing equably, without relation to anything external — and it is so deeply built into ordinary experience that it takes an effort to notice it is an assumption at all. Every other equation in (2.1.1) will survive into Chapter 2.2 in modified form. That one will not survive at all.

1.2 · Velocities add

A particle traces out a path. In S its position is $x(t)$; in S' , by (2.1.1), it is $x'(t') = x(t) - vt$. Differentiate with respect to t' — which, because $t' = t$, is the same operation as differentiating with respect to t :

$$u' \equiv \frac{dx'}{dt'} = \frac{d}{dt}(x - vt) = \frac{dx}{dt} - v = u - v. \quad (2.1.2)$$

Velocities subtract when you change frames, which is the same statement as: velocities *add* when you compose motions. A ball thrown forward at u' inside a train moving at v is seen from the platform to move at $u' + v$. Nobody has ever been surprised by this, and for material objects at ordinary speeds it is right to many decimal places.

Note precisely which step used which assumption. The $-v$ came from the spatial part of (2.1.1); the fact that we could differentiate "with respect to t' " and "with respect to t " interchangeably came from $t' = t$. Chapter 2.2 keeps the first ingredient and destroys the second, and (2.1.2) is what changes.

1.3 · Accelerations do not change, so $\mathbf{F} = m\mathbf{a}$ survives

Differentiate (2.1.2) once more. The relative velocity v is a constant — this is the definition of an inertial frame moving uniformly — so its derivative vanishes:

$$a' = \frac{du'}{dt'} = \frac{d}{dt}(u - v) = \frac{du}{dt} = a. \quad (2.1.3)$$

In three dimensions the same computation on each component gives $\ddot{\mathbf{r}}' = \ddot{\mathbf{r}}$. **Acceleration is a Galilean invariant.** Two observers in uniform relative motion disagree about where a particle is and about how fast it is going, and agree exactly about how it is accelerating.

That single fact is why Newtonian mechanics is compatible with the principle of relativity, and the argument runs as follows. Newton's second law in S reads $\mathbf{F} = m\ddot{\mathbf{r}}$. Transform it. The mass m is a property of the body and does not depend on who is watching. The acceleration is unchanged by (2.1.3). So the law in S' reads $\mathbf{F} = m\ddot{\mathbf{r}}'$, and the only remaining question is whether the *force* is the same in both frames.

For the forces Newtonian physics actually uses, it is. Every one of them depends on the particles' relative positions, and possibly on their relative velocities:

$$\begin{aligned} \mathbf{r}'_i - \mathbf{r}'_j &= (\mathbf{r}_i - \mathbf{v}t) - (\mathbf{r}_j - \mathbf{v}t) = \mathbf{r}_i - \mathbf{r}_j, \\ \dot{\mathbf{r}}'_i - \dot{\mathbf{r}}'_j &= (\dot{\mathbf{r}}_i - \mathbf{v}) - (\dot{\mathbf{r}}_j - \mathbf{v}) = \dot{\mathbf{r}}_i - \dot{\mathbf{r}}_j. \end{aligned} \quad (2.1.4)$$

The boost cancels in both differences. Gravity depends on $|\mathbf{r}_i - \mathbf{r}_j|$; Coulomb's law and Hooke's law likewise; even a drag force proportional to the relative velocity of body and fluid is untouched. Therefore $\mathbf{F}' = \mathbf{F}$ for each of them, and

$$\mathbf{F} = m\ddot{\mathbf{r}} \quad \implies \quad \mathbf{F}' = m\ddot{\mathbf{r}}'. \quad (2.1.5)$$

The law has the *same shape* in the new frame. Not merely a shape that can be corrected into the old one by adding terms — the identical shape, with the identical symbols meaning the identical things. Compare Chapter 1.1 §4.2, where writing $\mathbf{F} = m\mathbf{a}$ in polar coordinates produced two extra terms out of nothing. Nothing like that happens here.

▼ Grind box — the full Galilean group, and what "same form" is allowed to mean

(2.1.1) is one member of a larger family. The complete set of transformations that carry one inertial frame to another, in Newtonian physics, is

$$\mathbf{r}' = R\mathbf{r} - \mathbf{v}t + \mathbf{d}, \quad t' = t + s,$$

where R is a constant rotation matrix (3 parameters), $\mathbf{v}$ a constant boost velocity (3), $\mathbf{d}$ a constant spatial displacement (3) and s a constant time offset (1). Ten parameters. They compose and invert among themselves — the composition of two boosts is a boost with $\mathbf{v}_1 + \mathbf{v}_2$, the inverse of a rotation is a rotation — so they form a **group**, the **Galilean group**. Chapter 6.1 gives that word a definition; for now the content is just that "change of inertial frame" is a closed operation.

Differentiating twice kills $\mathbf{d}$ and $\mathbf{v}t$ outright and leaves $\ddot{\mathbf{r}}' = R\ddot{\mathbf{r}}$, so acceleration is not quite invariant under the full group — it rotates, as a vector must. But it rotates the *same way* $\mathbf{F}$ does, so $\mathbf{F} = m\ddot{\mathbf{r}}$ still maps to $\mathbf{F}' = m\ddot{\mathbf{r}}'$. That is the precise sense in which "the form is unchanged": **both sides of the equation transform identically, so the equality is preserved**. Hold onto that formulation. It is the exact criterion Chapter 2.4 turns into the definition of a tensor equation, and it is the reason index notation exists.

What is not allowed. Let the relative velocity depend on time, $\mathbf{r}' = \mathbf{r} - \mathbf{s}(t)$ with $\ddot{\mathbf{s}} \neq \mathbf{0}$. Then

$$\ddot{\mathbf{r}}' = \ddot{\mathbf{r}} - \ddot{\mathbf{s}}(t) \quad \implies \quad m\ddot{\mathbf{r}}' = \mathbf{F} - m\ddot{\mathbf{s}}(t),$$

and an extra term has appeared that is not any interaction between bodies. That is the fictitious force of Chapter 1.1 §1.1, and its presence is exactly why the first law is an existence claim about a restricted class of frames rather than a corollary of the second. The principle of relativity was never a claim about *all* frames. It is a claim about the inertial ones, and the boost parameter must be constant for the argument of §1.3 to go through.

One more piece of fine print, cheap now and expensive later. Nothing above proves that (2.1.1) is the *only* transformation with these properties. We derived $\mathbf{F}' = m\mathbf{a}'$ from the Galilean transformation; we did not show that a theory respecting the principle of relativity has to use it. That gap is the whole of Chapter 2.2, and the reason nobody looked into it for three hundred years is that $t' = t$ did not feel like an assumption.

1.4 · The principle itself

Now separate the two ideas that §1.1–1.3 ran together.

THE PRINCIPLE OF RELATIVITY

The laws of physics take the same form in all inertial frames. No experiment performed entirely within a uniformly moving laboratory can determine that laboratory's velocity.

This is not Einstein's. It is Galileo's, from 1632, and his statement of it is better than most modern ones: shut yourself below decks on a large ship with some flies, a bowl of water, and a friend to throw you a ball, and you will find that everything proceeds exactly as it did in port — the flies do not pile up at the stern, the ball needs no extra effort thrown forwards rather than backwards, the drops fall straight into the vessel beneath. ▀ (Paraphrased from the *Dialogo*; the reasoning is his, the words are not.) By 1900 nobody proposed to abandon it. It was, and is, one of the best-tested statements in physics.

The principle says something about the *relationship* between frames. It does not say what the transformation between them is. Those are separate questions, and the failure to notice they were separate is the single reason the crisis took forty years to resolve. Written out:

	CLAIM	STATUS IN 1900
(P)	The laws take the same form in all inertial frames	Universally accepted; verified constantly
(G)	Inertial frames are related by (2.1.1), with $t' = t$	Assumed without comment by everyone

Everybody held both. It had never occurred to anyone that they were different claims, because (G) was not perceived as a claim — it was perceived as what the words in (P) *meant*. The resolution of this chapter's crisis is that (P) is right and (G) is wrong. Keep the pair separate from here on; the rest of the chapter is about the strain between them.

2 · Maxwell's equations produce a speed

Now the second of the three claims, and it must be derived rather than quoted, because the number that falls out at the end is the whole point.

▮ QUOTED, NOT DERIVED — MAXWELL'S EQUATIONS

The four equations below are the empirical input of this chapter. Each summarises a body of nineteenth-century laboratory work — Coulomb, Gauss, Ørsted, Ampère, Faraday — and the last term in the last equation is Maxwell's own addition, which Chapter 0.7 Problem 4 showed is *forced* by charge conservation and the identity $\nabla \cdot (\nabla \times \mathbf{A}) = 0$. In vacuum ($\rho = 0, \mathbf{J} = \mathbf{0}$) and in SI units:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= 0, & \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \cdot \mathbf{B} &= 0, & \nabla \times \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.\end{aligned}$$

Nothing else about electromagnetism is assumed anywhere in this chapter. The two constants ϵ_0 and μ_0 are measured, and §2.4 says exactly how. Chapter 2.6 rewrites all four equations as a single manifestly Lorentz-covariant statement and shows that they were never four equations at all; Chapter 6.3 derives them from a symmetry principle. Here they are input.

2.1 · One identity first

The derivation needs the curl of a curl, which Chapter 0.7 §7 did not have occasion to compute. It is worth deriving rather than looking up, because the pattern of the answer is not guessable.

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}. \quad (2.1.6)$$

Here $\nabla^2 \mathbf{A}$ means the Laplacian of Chapter 0.7 §7.5 applied to each Cartesian component separately — a definition that is only available *because* the components are Cartesian, a caveat that becomes important in Chapter 3.3 and is harmless here.

▼ Grind box — (2.1.6) by brute force, one component

Write $\mathbf{C} = \nabla \times \mathbf{A}$, whose components are, from Chapter 0.7 §4,

$$C^x = \partial_y A^z - \partial_z A^y, \quad C^y = \partial_z A^x - \partial_x A^z, \quad C^z = \partial_x A^y - \partial_y A^x.$$

Take the x -component of $\nabla \times \mathbf{C}$ and substitute:

$$\begin{aligned}(\nabla \times \mathbf{C})^x &= \partial_y C^z - \partial_z C^y \\ &= \partial_y (\partial_x A^y - \partial_y A^x) - \partial_z (\partial_z A^x - \partial_x A^z) \\ &= \partial_y \partial_x A^y + \partial_z \partial_x A^z - \partial_y \partial_y A^x - \partial_z \partial_z A^x.\end{aligned}$$

Now the trick, and it is the only step with any content: *add and subtract the missing term* $\partial_x \partial_x A^x$. The first group then completes to a divergence and the second to a Laplacian:

$$(\nabla \times \mathbf{C})^x = \underbrace{\partial_x \partial_x A^x + \partial_y \partial_x A^y + \partial_z \partial_x A^z}_{\partial_x (\nabla \cdot \mathbf{A})} - \underbrace{\partial_x \partial_x A^x + \partial_y \partial_y A^x + \partial_z \partial_z A^x}_{\nabla^2 A^x}.$$

Both regroupings used Clairaut's theorem (Chapter 0.6 §6.1) to swap $\partial_y \partial_x$ for $\partial_x \partial_y$, whose hypothesis is continuous second partials — the same hypothesis that made $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ work in Chapter 0.7 §7.2. The y and z components follow by cycling $x \rightarrow y \rightarrow z \rightarrow x$, under which every expression above is invariant. Hence (2.1.6). ✓

Why the answer looks like that. $\nabla \times (\nabla \times \cdot)$ is a second-order operator, so it must be some combination of $\partial_i \partial_j$ acting on A^k . Only two such combinations produce a vector: the gradient of the divergence, and the Laplacian of the vector. The identity says the coefficients are $+1$ and -1 . In Chapter 3.5 this becomes the statement that the operator $d \star d$ splits in exactly this way, and the minus sign is the one that makes the electromagnetic wave equation come out with the right relative sign between space and time.

2.2 · Take the curl of Faraday's law

We want an equation for $\mathbf{E}$ alone. Faraday's law relates $\mathbf{E}$ to $\mathbf{B}$; Ampère–Maxwell relates $\mathbf{B}$ back to $\mathbf{E}$. So apply one to the other and eliminate $\mathbf{B}$. The operation that lets Ampère–Maxwell in is the curl, because the curl of $\mathbf{B}$ is precisely what that law tells us. Start from Faraday and hit both sides with $\nabla \times$:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}). \quad (2.1.7)$$

The second step swapped a space derivative with a time derivative. That is Clairaut again — the curl is built from $\partial_x, \partial_y, \partial_z$, none of which is ∂_t , so as long as $\mathbf{B}$ has continuous second partials the two operations commute. It is a small step and worth naming, because in Chapter 3.3 the analogous swap will *fail* and the failure will be the curvature of spacetime.

Now substitute Ampère–Maxwell on the right, $\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t$:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (2.1.8)$$

And expand the left with (2.1.6), using $\nabla \cdot \mathbf{E} = 0$ — which holds because we are in vacuum, with no charge anywhere to source a divergence:

$$\nabla \times (\nabla \times \mathbf{E}) = \underbrace{\nabla (\nabla \cdot \mathbf{E})}_{=0} - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}. \quad (2.1.9)$$

Set the two expressions equal and cancel the overall minus sign:

$$\boxed{\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}} \quad (2.1.10)$$

Look at what has happened. We started with four equations coupling two fields, performed three manipulations, and produced an equation for $\mathbf{E}$ by itself with no $\mathbf{B}$ in it. And the equation is not a new object: it is the wave equation of Chapter 0.8 §7.6,

$$\frac{\partial^2 u}{\partial t^2} = v^2 \frac{\partial^2 u}{\partial x^2}, \quad \text{here} \quad \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{1}{\mu_0 \epsilon_0} \nabla^2 \mathbf{E}, \quad (2.1.11)$$

with three spatial dimensions instead of one and a vector where the string had a scalar. Reading off the coefficient:

$$v^2 = \frac{1}{\mu_0 \epsilon_0} \quad \Longrightarrow \quad \boxed{c \equiv \frac{1}{\sqrt{\mu_0 \epsilon_0}}} \quad (2.1.12)$$

Nobody put a speed in. Two constants measured with capacitors and current-carrying wires went in; a propagation speed came out.

▼ Grind box — the same for $\mathbf{B}$, and the plane-wave solution

The magnetic field obeys the identical equation. Run the argument the other way. Take the curl of Ampère–Maxwell:

$$\nabla \times (\nabla \times \mathbf{B}) = \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{E}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2},$$

using Faraday in the last step. Expand the left with (2.1.6) and use $\nabla \cdot \mathbf{B} = 0$ — which, note, holds *always*, not merely in vacuum, since there are no magnetic charges. So

$$\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2},$$

the same equation with the same speed. The two fields propagate together, which they must — they are two aspects of one thing, as Chapter 2.6 makes literal.

A plane wave, and $c = \omega/k$. Try a solution travelling along x with $\mathbf{E}$ pointing along y : $\mathbf{E} = \hat{\mathbf{y}} E_0 \cos(kx - \omega t)$. Then $\nabla^2 \mathbf{E} = -k^2 \mathbf{E}$ and $\partial^2 \mathbf{E} / \partial t^2 = -\omega^2 \mathbf{E}$, so (2.1.10) requires

$$-k^2 = -\mu_0 \epsilon_0 \omega^2 \quad \Longrightarrow \quad \frac{\omega}{k} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c.$$

That is Chapter 0.8's relation between frequency, wavenumber and wave speed, now with the speed fixed by two laboratory constants. Notice that it holds for *every* k : the speed does not depend on the wavelength, so vacuum is non-dispersive and a pulse of any shape travels undistorted — which is

Chapter 0.8's $f(x - ct)$, and which is why we see sharp images of distant stars rather than smeared ones.

One honest caveat. (2.1.10) is a *consequence* of Maxwell's equations, not an equivalent of them. Every solution of Maxwell solves it; not every solution of it solves Maxwell. Feeding the plane wave back into the original four equations imposes two further conditions we did not need above — $\nabla \cdot \mathbf{E} = 0$ forces $\mathbf{E} \perp$ propagation direction (the wave is **transverse**), and Faraday fixes $\mathbf{B} = \hat{\mathbf{x}} \times \mathbf{E}/c$, so $\mathbf{B}$ is perpendicular to both $\mathbf{E}$ and the direction of travel, in phase with $\mathbf{E}$, and smaller by a factor c . None of that is needed for this chapter's argument, and all of it matters in Chapter 2.6.

2.3 • Put the numbers in

Now the moment. Here is where ϵ_0 and μ_0 come from, because the provenance is the point.

▀ ϵ_0 , **the permittivity of free space**, is fixed by measuring the capacitance of a parallel-plate capacitor: put a known charge on two plates of known area and separation, measure the voltage, and ϵ_0 falls out of $C = \epsilon_0 A/d$. This is a tabletop electrostatics experiment involving no light, no motion, and no waves. Its value is $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$.

▀ μ_0 , **the permeability of free space**, is fixed by measuring the force between two parallel current-carrying wires: run known currents through them, measure the attraction per unit length, and μ_0 falls out of $F/\ell = \mu_0 I_1 I_2 / 2\pi d$. This is a tabletop magnetostatics experiment involving no light and no waves either. Its value is $\mu_0 = 4\pi \times 10^{-7} \text{ N A}^{-2} = 1.2566 \times 10^{-6} \text{ N A}^{-2}$.

Multiply them together and take the reciprocal square root:

$$\begin{aligned} \mu_0 \epsilon_0 &= (1.2566 \times 10^{-6}) (8.854 \times 10^{-12}) = 1.1126 \times 10^{-17} \text{ s}^2 \text{ m}^{-2}, \\ c &= \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{3.3356 \times 10^{-9} \text{ s m}^{-1}} = 2.998 \times 10^8 \text{ m s}^{-1}. \end{aligned} \tag{2.1.13}$$

The measured speed of light, from Fizeau's toothed wheel in 1849 and Foucault's rotating mirror in 1862, was ▀ about $3.0 \times 10^8 \text{ m s}^{-1}$.

WHAT JUST HAPPENED

A capacitor and a pair of wires told you the speed of light.

Not "a number of the same order". The *speed of light*, to the precision of the input data, out of two constants that were measured in experiments containing no light at all. There is no route by which optics could have leaked into either measurement — one is a static charge on metal plates, the other a steady current in a wire. The only way the number can come out right is if light is the propagating disturbance (2.1.10) describes.

■ Maxwell drew exactly that conclusion in 1862, in one of the most consequential sentences in physics: *we can scarcely avoid the inference that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena*. Optics — a subject two thousand years old, with its own laws, its own instruments and its own practitioners — stopped being a separate science and became a chapter of electromagnetism. Hertz generated and detected the waves directly in 1887, and radio, X-rays and everything else followed from taking (2.1.12) seriously across the spectrum.

This is what a successful unification looks like from the inside: not a philosophical reorganisation, but two numbers you measured for unrelated reasons combining into a third number you had already measured for a third reason.

△ WHY THIS ISN'T OBVIOUS — IS (2.1.13) CIRCULAR?

A sharp reader will object. In the SI system in force for most of the twentieth century, μ_0 was *defined* to be exactly $4\pi \times 10^{-7}$, as part of the definition of the ampere; and since 1983 the metre has been defined by fixing c exactly. If the constants are defined rather than measured, has anything been discovered?

Yes, and the objection is worth taking seriously because it locates the physical content precisely. Unit conventions can move a numerical factor from one constant to another; they cannot create a relationship between independently measurable quantities. The invariant statement is this: **the ratio of the electrostatic to the electromagnetic unit of charge has the dimensions of a speed, and that speed can be measured with nothing but charges, currents and a balance**. Weber and Kohlrausch did precisely that in 1856 — discharging a capacitor of measured electrostatic capacity through a galvanometer of measured electromagnetic sensitivity — and got ■ $3.107 \times 10^8 \text{ m s}^{-1}$, six years before Maxwell's remark and with no reference to light whatever. That measurement is the physics. It cannot be conjured out of a choice of units, and it is the reason the coincidence was persuasive at the time.

In modern SI the bookkeeping runs the other way: c is exact by definition, μ_0 is now a measured quantity (differing from $4\pi \times 10^{-7}$ in the ninth decimal place), and (2.1.12) is the relation used to determine it. The equation is doing work in both directions; only the label "defined" has moved.

3 · The question nobody could answer: c with respect to what?

Now the trouble starts, and it starts with a question that is embarrassing in its simplicity.

(2.1.12) gives a speed. A speed is a rate of change of position, and position is measured relative to something. So: *relative to what is the light going at c ?*

3.1 · Every other wave answers this question easily

Chapter 0.8 §7.6 built the wave equation from a chain of masses and springs and got $v = \sqrt{T/\rho}$. Look at what that formula is made of: the tension and the linear density of the *string*. Both are properties of the medium, so the speed is a property of the medium, and it is obviously the speed *relative to the medium* — if the string is being reeled in while the wave travels along it, an observer at the side of the room measures something else.

Every wave in nineteenth-century physics was like this, without exception:

WAVE	SPEED	RELATIVE TO
Transverse wave on a string	$\sqrt{T/\rho}$	the string
Sound in air	$\sqrt{\gamma p/\rho} \approx 343 \text{ m s}^{-1}$	the air
Ripples on water	$\sqrt{g\lambda/2\pi}$ (deep water)	the water
Seismic S -waves	$\sqrt{G/\rho}$	the rock

And the consequence is entirely familiar. Stand in a wind of speed u and sound travelling downwind reaches you at $343 + u$, upwind at $343 - u$. That is (2.1.2) applied to a wave, and it is measured routinely. **The speed in the formula is the speed in the rest frame of the medium; in any other frame you add the medium's velocity.**

3.2 · The wave equation itself makes the point

You do not even need the physical picture. The mathematics says it directly. The wave equation is written in some particular set of coordinates (x, t) , and its solutions are $f(x - vt)$ and $g(x + vt)$ — disturbances moving at $\pm v$ in *those coordinates*. If the equation holds in one coordinate system with speed v , and if coordinates transform by (2.1.1), then in a coordinate system moving at u the same disturbance moves at $v - u$. The number v in the equation is therefore attached to one preferred frame — the frame in which the equation takes that form.

Apply this to (2.1.10). Maxwell's equations, written as they always are written, hold in *some* frame. In that frame light goes at c in every direction. In a frame moving at u through it, light should go at $c - u$ in one direction and $c + u$ in the other. So Maxwell's equations appear to single out a preferred frame of reference — which flatly contradicts claim (P) of §1.4, and does so for reasons that have nothing to do with any experiment.

3.3 · The ether, taken seriously

Given all of §3.1, there was one obvious move, and it was not a stupid one. If light is a wave, it is a wave *in something*. Name that something the **luminiferous ether**, declare Maxwell's equations to hold in the ether's rest frame, and the preferred frame of §3.2 is no longer an embarrassment — it is simply the rest frame of a medium, exactly like the air for sound. The principle of relativity survives untouched, because the ether is a physical object and detecting motion relative to it is no more mysterious than feeling a breeze.

This was not a fudge. It was the *only known way for a wave to work*, and every other wave anyone had ever studied confirmed it. Refusing to posit a medium in 1880 would have been the strange move, not the sober one. And the hypothesis was productive: it made a sharp prediction — the Earth moves, so there must be an ether wind, so the speed of light must be anisotropic in the laboratory — and that prediction is testable. §5 tests it.

The ether was, however, a demanding object. Its properties were fixed by (2.1.10) and they do not sit comfortably together.

▼ Grind box — what the ether had to be like, quantitatively

It had to be a solid. Light is transverse — a plane wave has **E** perpendicular to the direction of travel, as the previous grind box showed. A transverse wave is a shear disturbance: neighbouring layers of the medium slide past one another. Fluids do not resist shear, which is why sound in air is purely longitudinal and why the Earth's liquid outer core transmits no *S*-waves. So the ether had to have *rigidity*. It had to be an elastic solid filling all of space, and the planets had to pass through it without measurable resistance.

And an extraordinarily stiff one. For a transverse wave in an elastic solid the speed is $v = \sqrt{G/\rho}$ with G the shear modulus, the same square root as $\sqrt{T/\rho}$ in Chapter 0.8. Setting $v = c$ fixes not the stiffness and not the density but their ratio:

$$\frac{G}{\rho} = c^2 = 8.99 \times 10^{16} \text{ m}^2 \text{ s}^{-2}.$$

Compare steel: shear modulus 79.3 GPa, density 7850 kg m⁻³, so $G/\rho = 1.01 \times 10^7 \text{ m}^2 \text{ s}^{-2}$ — and indeed $\sqrt{1.01 \times 10^7} = 3.18 \text{ km s}^{-1}$, which is the shear-wave speed in steel. The ether's stiffness-to-density ratio therefore had to exceed steel's by a factor

$$\frac{8.99 \times 10^{16}}{1.01 \times 10^7} = 8.9 \times 10^9,$$

nearly ten billion. You may make the density as small as you like to keep the planets moving, but the ratio is not negotiable, and it is the ratio that has to be explained.

And it had to be undetectable in every other way. Transparent, frictionless, incompressible enough to fill the space between the stars, non-interacting with matter except through this one channel, and possessed of no measurable effect on anything except that light goes through it.

Nineteenth-century physicists were entirely aware of this list, and it bothered them; Kelvin called the ether one of the two clouds on the horizon of physics. But an awkward hypothesis that makes a testable prediction is a perfectly respectable scientific object, and the response was the right one: *go and measure the wind*.

4 · Maxwell's equations are not Galilean invariant

Section 3 argued from physical analogy that Maxwell's equations must pick out a frame. This section proves it, by computation, with no analogy anywhere. It is the hinge of the chapter.

Strip the problem to its skeleton. Take one Cartesian component of $\mathbf{E}$, call it ϕ , suppress y and z so that the Laplacian is $\partial^2/\partial x^2$, and (2.1.10) becomes the one-dimensional wave equation:

$$\frac{\partial^2 \phi}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2}. \quad (2.1.14)$$

The question: what does this equation look like in a frame moving at v along x ? Not what do its *solutions* look like — what does the *equation* look like. If the principle of relativity holds with the Galilean transformation, the answer must be $\partial^2 \phi / \partial x'^2 = c^{-2} \partial^2 \phi / \partial t'^2$, character for character.

4.1 · The chain rule, set up carefully

We have new coordinates $x' = x - vt$, $t' = t$, and a function ϕ that we may regard either as a function of (x, t) or of (x', t') — it is one physical field, described two ways. Chapter 0.6 §5.2 gives the rule for converting derivatives: each old derivative becomes a sum over the new ones, weighted by how each new coordinate responds to the old one.

$$\frac{\partial}{\partial x} = \frac{\partial x'}{\partial x} \frac{\partial}{\partial x'} + \frac{\partial t'}{\partial x} \frac{\partial}{\partial t'}, \quad \frac{\partial}{\partial t} = \frac{\partial x'}{\partial t} \frac{\partial}{\partial x'} + \frac{\partial t'}{\partial t} \frac{\partial}{\partial t'}. \quad (2.1.15)$$

The four partial derivatives of the new coordinates with respect to the old come straight from $x' = x - vt$ and $t' = t$:

$$\frac{\partial x'}{\partial x} = 1, \quad \frac{\partial x'}{\partial t} = -v, \quad \frac{\partial t'}{\partial x} = 0, \quad \frac{\partial t'}{\partial t} = 1. \quad (2.1.16)$$

Take a moment over $\partial t' / \partial x = 0$. It says: *changing where you are does not change what time it is*. That is the mathematical form of absolute simultaneity, it is the only place $t' = t$ enters this calculation, and it is the entry that Chapter 2.2 will make nonzero. Substituting (2.1.16) into (2.1.15):

$$\boxed{\frac{\partial}{\partial x} = \frac{\partial}{\partial x'}, \quad \frac{\partial}{\partial t} = \frac{\partial}{\partial t'} - v \frac{\partial}{\partial x'}}. \quad (2.1.17)$$

The spatial derivative is untouched; the time derivative acquires a piece. That is the whole asymmetry, and everything below is bookkeeping. The physical reading of the second relation is worth having: *holding your position fixed in the old frame means drifting backwards at $-v$ in the new one, so the rate of change you measure standing still picks up a term from the motion*. It is the convective derivative of fluid mechanics, arrived at without meaning to.

4.2 · Second derivatives, and the term that ruins everything

▼ Grind box — squaring the operators, every step

Space. Apply (2.1.17) twice:

$$\frac{\partial^2}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \right) = \frac{\partial}{\partial x'} \left(\frac{\partial}{\partial x'} \right) = \frac{\partial^2}{\partial x'^2}.$$

Nothing to do.

Time. Here we must square a sum of two operators, and the one thing to be careful about is that operators need not commute, so $(A + B)^2 = A^2 + AB + BA + B^2$ and we may *not* jump to $A^2 + 2AB + B^2$ without checking. Write $A = \partial/\partial t'$ and $B = -v \partial/\partial x'$:

$$\begin{aligned} \frac{\partial^2}{\partial t^2} &= \left(\frac{\partial}{\partial t'} - v \frac{\partial}{\partial x'} \right) \left(\frac{\partial}{\partial t'} - v \frac{\partial}{\partial x'} \right) \\ &= \frac{\partial^2}{\partial t'^2} - v \frac{\partial}{\partial t'} \frac{\partial}{\partial x'} - v \frac{\partial}{\partial x'} \frac{\partial}{\partial t'} + v^2 \frac{\partial^2}{\partial x'^2}. \end{aligned}$$

The two middle terms are equal *because of Clairaut's theorem* — mixed partials of a twice-continuously-differentiable function commute — and because v is a constant, so it passes through $\partial/\partial x'$ without generating anything. Both facts are needed, and both are easy to use without noticing. Hence

$$\frac{\partial^2}{\partial t^2} = \frac{\partial^2}{\partial t'^2} - 2v \frac{\partial^2}{\partial x' \partial t'} + v^2 \frac{\partial^2}{\partial x'^2}.$$

Assemble. Substitute both results into (2.1.14):

$$\frac{\partial^2 \phi}{\partial x'^2} = \frac{1}{c^2} \left[\frac{\partial^2 \phi}{\partial t'^2} - 2v \frac{\partial^2 \phi}{\partial x' \partial t'} + v^2 \frac{\partial^2 \phi}{\partial x'^2} \right].$$

Bring the v^2 term to the left and collect the coefficient of $\partial^2 \phi / \partial x'^2$:

$$\left(1 - \frac{v^2}{c^2} \right) \frac{\partial^2 \phi}{\partial x'^2} + \frac{2v}{c^2} \frac{\partial^2 \phi}{\partial x' \partial t'} = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t'^2},$$

which is (2.1.18) below, rearranged.

The result of that grind, written with everything on one side and $\beta \equiv v/c$ as the convention of Part II requires:

$$(1 - \beta^2) \frac{\partial^2 \phi}{\partial x'^2} + \frac{2v}{c^2} \frac{\partial^2 \phi}{\partial x' \partial t'} - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t'^2} = 0. \quad (2.1.18)$$

Compare with what we wanted, which was (2.1.14) with primes. Two things have gone wrong, and they are worth naming separately.

(i) A cross term has appeared. The middle term $\propto \partial^2 \phi / \partial x' \partial t'$ has no counterpart in (2.1.14). It is odd in v — reverse the direction of motion and it changes sign — which is precisely how an equation encodes a preferred direction. A term that knows which way is "forwards" cannot appear in an isotropic law.

(ii) The leading coefficient has changed. 1 has become $1 - \beta^2$. Even if you could somehow dispose of the cross term, the coefficients no longer match.

THE SHARP STATEMENT

The wave equation changes form under a Galilean boost. If it holds in one inertial frame, it does not hold in any other.

This is not an approximation, not a small effect, and not a subtlety of interpretation. It is a three-line consequence of the chain rule. And since (2.1.14) is a consequence of Maxwell's equations, the same is true of them: *Maxwell's equations are not Galilean invariant.*

4.3 · What the transformed equation is actually saying

(2.1.18) is not wrong. It is the correct description of a wave in a medium, as seen by an observer moving through that medium — and it is worth extracting exactly what it says, both to check the algebra and to see what is at stake.

Look for travelling-wave solutions in the new frame: $\phi = f(x' - ut')$ for some speed u and arbitrary shape f , exactly as in Chapter 0.8 Problem 4. Each derivative brings down a factor: $\partial_{x'}^2 \phi = f''$, $\partial_{x'} \partial_{t'} \phi = -u f''$, and $\partial_{t'}^2 \phi = u^2 f''$. Substituting into (2.1.18) and cancelling the common f'' (which is nonzero for any wave worth the name) gives a condition on u alone:

$$(1 - \beta^2) - \frac{2vu}{c^2} - \frac{u^2}{c^2} = 0 \quad \Longleftrightarrow \quad u^2 + 2vu - (c^2 - v^2) = 0. \quad (2.1.19)$$

A quadratic. Complete the square, or use the formula:

$$\begin{aligned} u &= -v \pm \sqrt{v^2 + c^2 - v^2} = -v \pm c \\ \implies u &= c - v \quad \text{or} \quad u = -(c + v). \end{aligned} \quad (2.1.20)$$

Exactly the two speeds §3.1 predicted on physical grounds: in the moving frame, light chases off in the $+x'$ direction at $c - v$ and comes back the other way at $c + v$. The algebra of §4.2 and the sound-in-a-wind picture of §3.1 are the same

statement.

So (2.1.18) is perfectly consistent physics — for a wave in a medium. Sound obeys the analogous equation and nobody minds, because sound has a medium and its rest frame is not mysterious. The entire question is whether light is like that.

THE FORK

We now have a genuine contradiction between two things both believed in 1900, and there are exactly three ways out. They are mutually exclusive and they are all uncomfortable.

(A) The principle of relativity fails for electromagnetism. Mechanics obeys it, optics does not; there is a preferred frame — the ether's — and Maxwell's equations hold only there. Every other frame gets (2.1.18). This was the majority view, it is entirely coherent, and it makes a prediction: *you can measure your velocity through the ether by measuring the speed of light in different directions.*

(B) Maxwell's equations are wrong, or at least are only the low-velocity limit of some Galilean-invariant electrodynamics still to be found. Several people tried; the attempts (Hertz's, Ritz's) all made predictions that failed. And they were fighting uphill, since Maxwell's equations were passing every test anyone could devise.

(C) The Galilean transformation is wrong. Frames are related by something else, and (2.1.1) is only its small- v approximation. Then the principle of relativity can hold for *both* mechanics and electromagnetism — but Newton's laws, which are exactly Galilean invariant by §1.3, must be modified instead.

Notice that (C) requires giving up $t' = t$. Nothing else in (2.1.1) has enough freedom in it. That is why (C) looked, in 1900, like the least attractive option available, and it is why the experiments came first.

5 • Michelson–Morley

Branch (A) is the one that is directly testable, and testing it is the most famous experiment in physics. The logic is simple: the Earth orbits the Sun at about 30 km s^{-1} , so unless the ether happens to be dragged along perfectly, there is an ether wind through the laboratory. Measure the speed of light along the wind and across it, and the two answers should differ.

The difficulty is that they differ by very little. From §4.3 the two one-way speeds are $c \pm v$ with $v/c \approx 10^{-4}$, so a direct timing measurement would have to resolve one part in 10^4 of a quantity that in 1887 was known to about one part in 10^3 . Michelson's answer was to stop measuring speed and start measuring *interference*, which compares two light paths against each other and can resolve a hundredth of a wavelength.

5.1 · The instrument

A source of monochromatic light shines on a half-silvered mirror — a **beam splitter** — set at 45° . Half the light passes through and continues to a mirror at distance L_1 ; half reflects off at right angles and travels to a second mirror at distance L_2 . Both beams return to the splitter, recombine, and go on to a detector. Whether they arrive in phase or out of phase depends on the difference in their travel times, and the recombined beam therefore shows a pattern of interference fringes.

Two facts about this arrangement do the real work. First, the arms are perpendicular, so if there is an ether wind one arm can be along it and the other across it. Second — and this is the piece of experimental cunning without which the whole thing is impossible — *you cannot know L_1 and L_2 to a fraction of a wavelength*, so the absolute fringe position is uninterpretable. What you can do is **rotate the whole apparatus by 90°** , which exchanges the roles of the two arms without changing either length, and watch the fringes move. §5.5 shows that the unknown mismatch $L_1 - L_2$ cancels out of that measurement exactly.

5.2 · The arm along the wind

Work in the ether's rest frame, where light travels at c in all directions and the apparatus moves at v in the $+x$ direction. (Working in the ether frame is the honest choice: it is the frame where we are entitled to say light moves at c , since that is the hypothesis under test.)

Set $t = 0$ when a pulse leaves the splitter, and put the splitter at the origin at that instant. Thereafter the splitter is at vt and the far mirror at $L_1 + vt$, both sliding to the right at v .

Outbound. The pulse moves right at c , so it is at ct ; the mirror is at $L_1 + vt$. They meet when

$$ct_{\text{out}} = L_1 + vt_{\text{out}} \quad \implies \quad t_{\text{out}} = \frac{L_1}{c - v}. \quad (2.1.21)$$

The mirror is running away from the light, so the light closes the gap at only $c - v$. Notice this is not a claim that the light moves at $c - v$ — it moves at c , in the ether frame, as assumed. The $c - v$ is a *closing speed* between two objects, and closing speeds may perfectly well exceed c or fall below it without anything moving at that rate.

Return. Now the light travels left at c from the mirror's position while the splitter advances to meet it. By the identical argument the gap closes at $c + v$:

$$t_{\text{back}} = \frac{L_1}{c + v}. \quad (2.1.22)$$

Add them, put over a common denominator, and divide top and bottom by c^2 :

$$\begin{aligned} t_{\parallel} &= \frac{L_1}{c - v} + \frac{L_1}{c + v} = \frac{L_1(c + v) + L_1(c - v)}{(c - v)(c + v)} = \frac{2L_1c}{c^2 - v^2} \\ &= \frac{2L_1/c}{1 - \beta^2}. \end{aligned} \quad (2.1.23)$$

The round trip is *slower* than the $2L_1/c$ it would take with no wind, by a factor $1/(1 - \beta^2)$. That is worth a sentence, because the intuition points the wrong way: the two legs seem as though they ought to average out, one gaining what the other loses. They do not, and the reason is that the leg travelled at the lower speed *lasts longer*, so it gets more weight in the total. Losing time at $c - v$ for a long while beats gaining it at $c + v$ briefly. The same asymmetry is why a round trip against and with a current always takes longer than still water, and why your average speed on a there-and-back drive is the harmonic mean and not the arithmetic one.

5.3 · The arm across the wind, done properly

This is where most treatments go soft, so we will be slow. The naive answer — the light goes straight across at c , so the round trip is $2L_2/c$ and the transverse arm is unaffected — is wrong, and it is wrong for a reason that becomes central in Chapter 2.2.

The mirror at the end of the transverse arm is *moving sideways*. During the time the light is in flight, that mirror slides a distance vt down the x -axis. Light that left the splitter travelling straight along y in the ether frame would arrive where the mirror used to be and miss it. To hit the mirror, the light must leave with a velocity that has an x -component matching the apparatus's own motion.

Write that as an equation. Let the light's velocity in the ether frame be $\mathbf{u} = (u_x, u_y)$ with, by hypothesis, $|\mathbf{u}| = c$. For the light to stay over the moving arm it needs $u_x = v$. Then

$$u_x^2 + u_y^2 = c^2, \quad u_x = v \quad \implies \quad u_y = \sqrt{c^2 - v^2}. \quad (2.1.24)$$

The *useful* component of the light's velocity — the part that gets it across the arm — is not c but $\sqrt{c^2 - v^2}$. The rest of its speed budget is spent keeping up sideways. Since the arm is L_2 long and the crossing is covered at u_y ,

$$t_{\perp} = \frac{2L_2}{\sqrt{c^2 - v^2}} = \frac{2L_2/c}{\sqrt{1 - \beta^2}}, \quad (2.1.25)$$

the factor of 2 because the return trip is the mirror image of the outbound one and takes the same time.

The same result comes out of a right triangle, which is the version to carry in your head. In time t_1 the light travels a straight-line distance ct_1 in the ether frame — the hypotenuse. The apparatus has drifted vt_1 — the base. The arm is L_2 — the height. Pythagoras:

$$\begin{aligned} (ct_1)^2 &= L_2^2 + (vt_1)^2 &\implies & t_1^2(c^2 - v^2) = L_2^2 \\ & &\implies & t_1 = \frac{L_2}{\sqrt{c^2 - v^2}}, \end{aligned} \quad (2.1.26)$$

and $t_{\perp} = 2t_1$ as before. Same answer, same square root, and the square root has the same origin in both derivations: **a fixed total speed c has to be shared between "across" and "along", and Pythagoras charges you for the sharing.** That triangle is going to reappear in Chapter 2.2 as the derivation of time dilation, with the arm replaced by a light clock, and the factor $1/\sqrt{1 - \beta^2}$ will be γ .

△ WHY THIS ISN'T OBVIOUS — THE TRANSVERSE ARM IS AFFECTED TOO

It is very natural to think the transverse arm is a clean reference: the wind blows sideways across it, so surely it is untouched, and all the physics is in the parallel arm. That reading is wrong, and getting it wrong changes the answer by a factor of two.

Compare the two results at small β , using the binomial series of Chapter 0.3:

$$t_{\parallel} = \frac{2L}{c} \left(1 + \beta^2 + \dots \right), \quad t_{\perp} = \frac{2L}{c} \left(1 + \frac{1}{2}\beta^2 + \dots \right).$$

Both arms are slowed. The parallel arm is slowed by γ^2 , the transverse arm by γ — where $\gamma = (1 - \beta^2)^{-1/2}$, so $\gamma^2 = (1 - \beta^2)^{-1}$ and the two exponents are exactly the ones sitting in (2.1.23) and (2.1.25). If you wrongly take $t_{\perp} = 2L/c$ you get a β^2 coefficient of 1 instead of $\frac{1}{2}$, and predict twice the fringe shift.

And the difference between γ and γ^2 is not an incidental detail — it is *the entire observable*. The experiment cannot measure either round-trip time; it can only compare them. Everything Michelson and Morley could possibly have seen lives in the gap between those two exponents, and if the two arms had been affected identically the experiment would have had nothing to detect no matter how sensitive it was. Chapter 2.2's resolution has to reproduce exactly this structure — one power of γ along the motion, none across it — and §6.2 shows that FitzGerald spotted the shape of the fix from precisely this observation, fifteen years before anyone knew why it was true.

5.4 · The difference

Take equal arms, $L_1 = L_2 = L$, which is what the instrument is built to have and what §5.5 shows we do not actually need. Subtract (2.1.25) from (2.1.23):

$$\Delta t \equiv t_{\parallel} - t_{\perp} = \frac{2L}{c} \left(\frac{1}{1 - \beta^2} - \frac{1}{\sqrt{1 - \beta^2}} \right). \quad (2.1.27)$$

That is exact. Now expand for small β , since $\beta \approx 10^{-4}$ and we want to know how big the effect is. Chapter 0.3's binomial series gives, with $s = \beta^2$,

$$\frac{1}{1 - s} = 1 + s + s^2 + \dots, \quad \frac{1}{\sqrt{1 - s}} = 1 + \frac{1}{2}s + \frac{3}{8}s^2 + \dots, \quad (2.1.28)$$

so the bracket in (2.1.27) is

$$\begin{aligned} (1 + s + s^2) - (1 + \frac{1}{2}s + \frac{3}{8}s^2) + \dots &= \frac{1}{2}s + \frac{5}{8}s^2 + \dots \\ &= \frac{1}{2}\beta^2 \left(1 + \frac{5}{4}\beta^2 + \dots \right), \end{aligned} \quad (2.1.29)$$

and therefore

$$\Delta t \approx \frac{2L}{c} \cdot \frac{\beta^2}{2} = \frac{L\beta^2}{c} = \frac{Lv^2}{c^3}. \quad (2.1.30)$$

The leading term vanishes: the two arms agree at order β^0 and at order β^1 , and first differ at order β^2 . **The effect is second order in v/c , and that is the central practical fact about the entire subject.** There is no first-order ether-wind effect in a round-trip measurement, because the delay on the way out is compensated to first order by the gain on the way back. Every experiment sensitive only to first order — and there had been several — was guaranteed a null result before it was built. Michelson's design was the first with second-order sensitivity, which is why it counts.

And that $\beta^2 = v^2/c^2$ is the same suppression factor that made the third law's failure so hard to notice in Chapter 1.1 §3.3. Both are the leading symptom of the same disease.

5.5 • What rotating the apparatus measures

Now the experimental point flagged in §5.1. You cannot measure Δt : it would require knowing $L_1 - L_2$ to a fraction of 590 nm, which nobody can build. So rotate.

Before rotation, with arm 1 along the wind and arm 2 across it, the time difference between the beams is, from (2.1.23) and (2.1.25),

$$\Delta t_{\text{before}} = \frac{2L_1/c}{1 - \beta^2} - \frac{2L_2/c}{\sqrt{1 - \beta^2}}. \quad (2.1.31)$$

Turn the table through 90° . Arm 1 is now across the wind and arm 2 along it; neither length has changed. So

$$\Delta t_{\text{after}} = \frac{2L_1/c}{\sqrt{1 - \beta^2}} - \frac{2L_2/c}{1 - \beta^2}. \quad (2.1.32)$$

Subtract. The terms regroup by *denominator* rather than by arm, and the arms enter only through their sum:

$$\begin{aligned} \Delta t_{\text{before}} - \Delta t_{\text{after}} &= \frac{2(L_1 + L_2)/c}{1 - \beta^2} - \frac{2(L_1 + L_2)/c}{\sqrt{1 - \beta^2}} \\ &= \frac{2(L_1 + L_2)}{c} \left(\frac{1}{1 - \beta^2} - \frac{1}{\sqrt{1 - \beta^2}} \right). \end{aligned} \quad (2.1.33)$$

The unknown mismatch $L_1 - L_2$ has vanished identically. Only $L_1 + L_2$ survives, and that is a quantity you can measure with a ruler. Setting $L_1 = L_2 = L$ so that $L_1 + L_2 = 2L$ and comparing with (2.1.27), the change on rotation is exactly $2\Delta t$.

A time difference τ between the two beams corresponds to an optical path difference $c\tau$, and one whole fringe of movement corresponds to a path difference of one wavelength λ . So the number of fringes the pattern shifts as the apparatus turns is

$$\Delta N = \frac{c \cdot 2\Delta t}{\lambda} = \frac{4L}{\lambda} \left(\frac{1}{1 - \beta^2} - \frac{1}{\sqrt{1 - \beta^2}} \right) \approx \frac{2L\beta^2}{\lambda}. \quad (2.1.34)$$

Clean, and made of measurable things: an arm length, a wavelength, and the square of the Earth's speed in units of c .

5.6 · The numbers, 1887

Michelson and Morley floated a sandstone slab on a bath of mercury so it could be turned smoothly without flexing, and folded each light path back and forth between multiple mirrors to lengthen it. ■ The effective one-way path was $L \approx 11 \text{ m}$; the source was a sodium lamp, $\lambda \approx 590 \text{ nm}$; and the assumed velocity was the Earth's orbital speed, $v = 30 \text{ km s}^{-1}$. Then

$$\beta = \frac{3.0 \times 10^4}{2.99792 \times 10^8} = 1.0007 \times 10^{-4}, \quad \beta^2 = 1.0014 \times 10^{-8}, \quad (2.1.35)$$

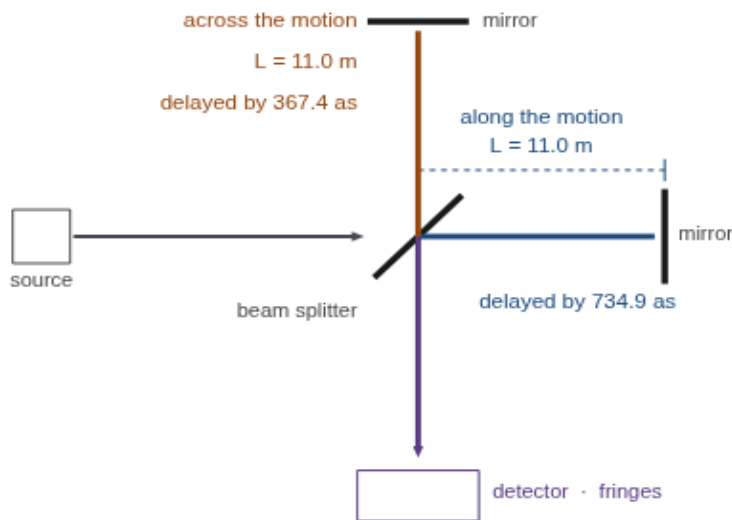
$$\Delta N \approx \frac{2(11 \text{ m})(1.0014 \times 10^{-8})}{5.90 \times 10^{-7} \text{ m}} = 0.373 \text{ fringes}. \quad (2.1.36)$$

(The exact expression (2.1.34) gives 0.373398; the expansion gives 0.373398. At $\beta = 10^{-4}$ the correction term $\frac{5}{4}\beta^2$ of (2.1.29) is one part in 10^8 , so the two agree to six figures, as they should.)

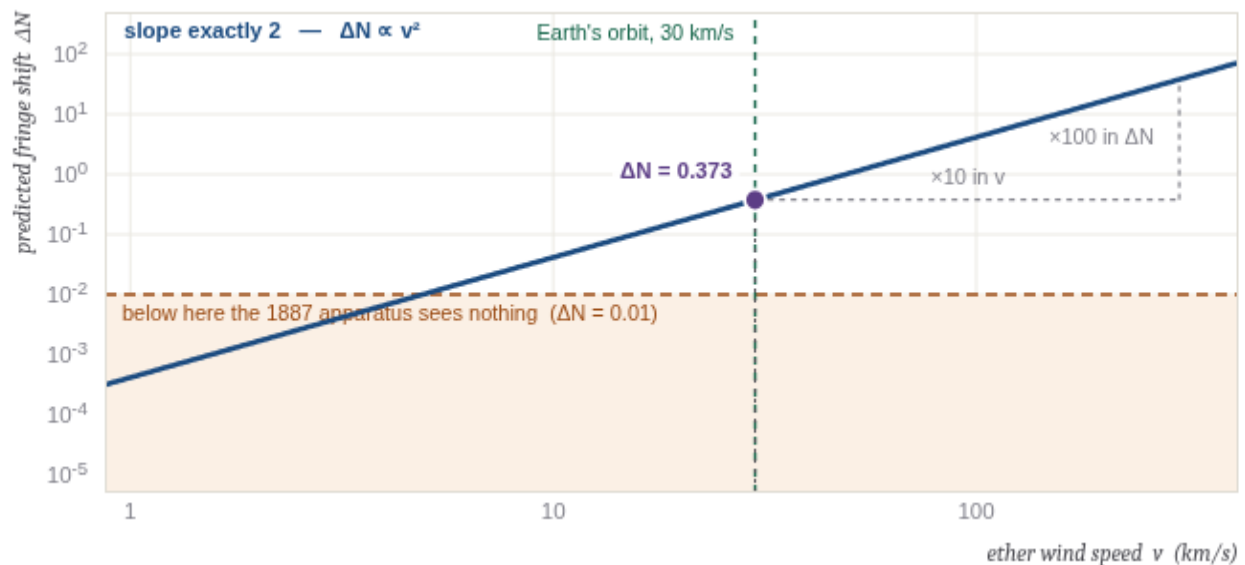
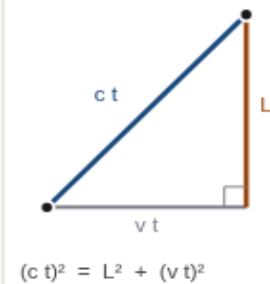
About four tenths of a fringe. Not enormous — but Michelson's interferometer could see a shift of about a hundredth of a fringe, so the predicted effect was roughly **forty times the detection threshold**.

5.7 · The interactive

the apparatus moves through the ether at $v = 30.0 \text{ km/s}$



the transverse crossing
(sideways drift hugely exaggerated)



ether wind v $v = 30.00 \text{ km/s}$ arm L $L = 11.0 \text{ m}$

Earth's orbital speed

1887 apparatus

$t_{\parallel} = 73.384101678 \text{ ns}$

$t_{\perp} = 73.384101311 \text{ ns}$

$\Delta t = 3.6743\text{e-}16 \text{ s} = 367.4 \text{ as}$

$\Delta N = 0.3734 \text{ fringes}$

$\Delta N / \text{bound} = 37.3\times$ – far above the threshold; crosses the bound at $v = 4.91 \text{ km/s}$

What the ether wind should have done, against what the experiment could see. The upper panel is the interferometer: source, beam splitter, the arm along the wind (blue) and the arm across it (orange), and the detector where the beams recombine. The inset at the right is the triangle of (2.1.26), drawn hugely out of scale — at the real $\beta \approx 10^{-4}$ the slant would be invisible — showing why the transverse crossing takes $L/\sqrt{c^2 - v^2}$ and not L/c .

The lower panel plots the predicted fringe shift against wind speed on **log–log axes**, where the β^2 scaling of (2.1.34) shows up as a straight line of slope exactly 2 — double the wind speed, quadruple the shift. The shaded band is the 1887 experimental upper bound of about 0.01 fringes; the dashed vertical line is the Earth's orbital speed of 30 km s^{-1} , which is not adjustable by anybody.

Drag the wind speed down and watch where the curve crosses into the band. At the 1887 apparatus ($L = 11 \text{ m}$) you have to get below about 4.9 km s^{-1} — a sixth of the Earth's orbital speed — before the prediction hides under the detection threshold. At the actual orbital speed the prediction sits a factor of 37 above it. **This is not a marginal call.** Push the arm length up and it gets worse for the ether, since ΔN is linear in L ; that is exactly why Michelson folded the path.

Every number is computed from the exact expressions (2.1.23), (2.1.25) and (2.1.34), never from the small- β expansion. Because $t_{\parallel}$ and $t_{\perp}$ agree to nine significant figures at realistic β , subtracting them directly in floating point would throw away most of the answer, so Δt is evaluated from the algebraically identical but numerically stable rearrangement $\Delta t = \frac{2L}{c} \beta^2 / [(1 - \beta^2)(1 + \sqrt{1 - \beta^2})]$ — which you can check reduces to (2.1.30) as $\beta \rightarrow 0$.

5.8 • The result

They found nothing.

■ Michelson and Morley reported in 1887 that the observed displacement was certainly less than a twentieth of the predicted 0.4 fringes, and probably less than a fortieth — an upper bound of about 0.01 fringes, which is to say the noise floor of the instrument. Repeated at different times of day and different seasons, to catch the Earth's rotation and its orbital motion pointing the apparatus in every possible direction relative to any conceivable ether frame. Nothing.

THIS IS NOT A SMALL DISCREPANCY

A great many experiments in physics disagree with prediction by twenty or thirty per cent and are eventually reconciled by a systematic effect somebody had missed. This is not one of them.

The prediction was 0.373 fringes. The bound was 0.01. That is not a mismatch in a coefficient — it is the **total absence of an effect that should have been unmissable**, by a factor of nearly forty, in an apparatus specifically designed to have several times the required sensitivity. Nor can it be rescued by supposing the Earth happens to be at rest in the ether at the moment of measurement: six months later the Earth's orbital velocity has reversed, so its speed relative to any fixed ether frame must then be at least 60 km s^{-1} and the predicted shift four times larger. Michelson and Morley looked. Nothing.

■ Modern versions replace the arms with cryogenic optical resonators and look for a directional dependence of the resonant frequency as the apparatus turns on a rotating table. The current bound on the fractional anisotropy of c is below one part in 10^{17} , some seven orders of magnitude tighter than what the 1887 apparatus could have detected. In a hundred and forty years of increasingly ferocious effort, the effect has not appeared.

6 · The rescues, and why they failed

A null result does not by itself dispose of the ether. It disposes of the *simplest* ether — one at rest in some frame, through which the Earth moves freely and which does nothing to the apparatus. Several serious alternatives were proposed, and it is worth being specific about them, because they were good physics and because the last one was very nearly right.

6.1 · Ether drag

The obvious first move: suppose the Earth drags the local ether with it, so that near the surface there is no wind at all. The interferometer sits in still air, so to speak, and of course sees nothing. Two independent observations rule this out.

Stellar aberration. Point a telescope at a star directly overhead. Because the Earth is moving sideways at v while the light travels down the tube, the telescope must be tilted slightly into the direction of motion — otherwise the eyepiece has moved out from under the light by the time it arrives. If the tube has length ℓ , the light takes ℓ/c to traverse it and the tube advances $v\ell/c$ in that time, so the required tilt is

$$\tan \theta = \frac{v\ell/c}{\ell} = \frac{v}{c} = \beta \quad \implies \quad \theta \approx \beta. \quad (2.1.37)$$

With $v = 29.78 \text{ km s}^{-1}$ this is $\theta = 9.933 \times 10^{-5} \text{ rad}$, and converting to arcseconds by multiplying by 206265,

$$\theta = 20.5''. \quad (2.1.38)$$

► Bradley discovered exactly this in 1728: every star traces out a small ellipse over the course of a year, of angular radius $20.4955''$, and the effect is the single most reliable confirmation that the Earth moves. Now put it together with drag. If the ether were carried along with the Earth, the light would enter that co-moving ether at the top of the atmosphere and thereafter travel in the telescope's own rest frame — and there would be no aberration at all, or at most a fringe effect at the boundary. The aberration is observed, it has exactly the size v/c predicted for undragged ether, and it is the same for all stars regardless of distance. Complete drag is dead.

Fizeau's water tube. In 1851 Fizeau sent light through a tube of water flowing at speed u and measured how much the water carried the light along with it. Complete drag predicts the light speed in the lab is $c/n + u$; no drag predicts c/n . He measured neither. ► The light was dragged by a *fraction* of the water's speed,

$$v_{\text{light}} = \frac{c}{n} + f u, \quad f = 1 - \frac{1}{n^2}, \quad (2.1.39)$$

which for water ($n = 1.333$) is $f = 0.437$, and Fizeau measured about 0.44. A partial drag coefficient is a very strange thing for a mechanical medium to have — how does a substance know to transmit exactly 43.7% of its own motion? — and no ether model ever gave a convincing account of it. (Chapter 2.2 will get (2.1.39) in one line as the leading term of relativistic velocity addition, with no medium, no drag, and no free parameter. It is one of the cleanest confirmations of the whole framework, and it is a nineteenth-century measurement.)

6.2 · The FitzGerald–Lorentz contraction

Now the interesting one. In 1889 FitzGerald, and independently Lorentz in 1892, made a proposal of startling economy: **objects moving through the ether contract along their direction of motion by exactly the factor $\sqrt{1 - \beta^2}$.**

Watch what that does. The parallel arm is along the motion, so its length in the ether frame is not L but $L\sqrt{1 - \beta^2}$. The transverse arm is perpendicular and is unaffected. Substitute into (2.1.23):

$$t_{\parallel} = \frac{2(L\sqrt{1 - \beta^2})/c}{1 - \beta^2} = \frac{2L/c}{\sqrt{1 - \beta^2}} = t_{\perp}. \quad (2.1.40)$$

The two round-trip times are equal. Not approximately equal, not equal to order β^2 — *identically equal*, for every β , because $\sqrt{1 - \beta^2}/(1 - \beta^2) = 1/\sqrt{1 - \beta^2}$ is an algebraic identity. The predicted fringe shift is exactly zero to all orders, and Michelson–Morley is explained.

It is impossible to look at (2.1.40) and not feel that something is being got away with. The contraction factor was chosen for no reason except that it makes the answer come out; it was, in FitzGerald's own framing, a hypothesis with one job. And yet it is not absurd. Lorentz gave it a physical rationale: if the forces holding matter together are electromagnetic in origin — which by 1892 was a reasonable guess — and if electromagnetic fields are distorted by motion through the ether, then the equilibrium spacing of the atoms in a solid might well change when the solid is set in motion. On that reading the contraction is a real, dynamical effect on *matter*, produced by the ether acting on the electrons in the rod. The rod is genuinely shorter. The ether is genuinely there. It simply conspires to be undetectable.

△ WHY THIS ISN'T OBVIOUS — A NULL RESULT DOES NOT REFUTE THE ETHER

The tidy story is that Michelson–Morley disproved the ether and Einstein cleared away the wreckage. Both halves are false, and the second is the more misleading.

(2.1.40) is a proof that a null result cannot kill a sufficiently determined theory. Lorentz's ether, equipped with contraction and (§6.3) local time, predicts *exactly* what Michelson and Morley saw — not approximately, exactly — and by 1904 it reproduced every optical experiment then known. No measurement of the kind being discussed could have distinguished it from what replaced it, because by construction the two make the same predictions.

What an experiment refutes is a *specific* model, not a concept. It refuted the undragged, undistorting ether. It said nothing about ethers that contract their occupants.

So what did kill it? Not evidence. **The ether stopped doing any work.** Every quantity it was introduced to explain — the propagation speed, the failure of the wind to show up, the aberration, the Fizeau coefficient — turned out to be calculable without it, and every attempt to detect it produced a null result that had to be patched by a new property whose only content was the null result it was patching. A frame of reference that no experiment can identify is not a frame of reference; it is a spare wheel bolted to the theory, and the moment you notice that removing it changes no prediction, it has been refuted in the only sense that matters.

That is a methodological point rather than an empirical one, and it deserves to be made without condescension. The nineteenth century was not being obtuse. They were doing exactly what you should do with a productive hypothesis under pressure: modify it minimally and see whether it survives. It did survive, technically, and it was abandoned anyway — which tells you that "consistent with all the data" is not the only standard a theory is held to. Compare the situation honestly with the present: Chapter 7.8 will ask the same question about ideas in quantum gravity that are consistent with everything and predict nothing, and the answer will be the same one, and it will be no more comfortable.

6.3 · Local time

Contraction alone is not quite enough, and Lorentz knew it. To get Maxwell's equations to come out with the same form in a moving frame — to make the theory work at first order in β for optical experiments generally, not just for Michelson–Morley — he found that he had to introduce a further substitution, which in his 1895 form reads

$$t' = t - \frac{vx}{c^2}. \quad (2.1.41)$$

He called t' the **local time** (*Ortszeit*), and he was explicit that he regarded it as a mathematical convenience with no physical meaning whatever. The *real* time was t , the ether's time, universal and absolute in Newton's sense. The quantity t' was an auxiliary variable — a change of integration variable that made the equations tractable, no more significant than substituting $u = x^2$ in an integral.

By 1904 Lorentz had the full set. Written in modern notation, with $\gamma = (1 - \beta^2)^{-1/2}$:

$$x' = \gamma(x - vt), \quad y' = y, \quad z' = z, \quad t' = \gamma\left(t - \frac{vx}{c^2}\right). \quad (2.1.42)$$

Those are the Lorentz transformations. Chapter 2.2 derives them from scratch and will not use this equation; it is here only so that the historical point can be made without ambiguity.

WHAT EINSTEIN ACTUALLY CONTRIBUTED

Lorentz had (2.1.42) before Einstein. Poincaré had noticed that the transformations form a group and had, in 1904, stated a principle of relativity covering electrodynamics. The algebra of special relativity was on paper, published, and being used.

So it is worth being exact about what 1905 added, because "Einstein derived the Lorentz transformation" is not it.

For Lorentz, x' and t' were *auxiliary variables*. There was a true frame — the ether's — with true coordinates x, t . Moving rods were really shortened, by a dynamical action of the ether on the electrical forces inside matter; moving clocks were not slowed at all, since t' was not a time but a substitution. The transformation described **what happens to matter when it moves through the ether**. It was a theory of rods and clocks.

For Einstein, x' and t' are *the coordinates a moving observer actually measures*, obtained from two postulates about the symmetry of nature and nothing else. No ether appears in the derivation because none is needed. There is no true frame, no true time, and nothing dynamical happens to the rod: the rod is shorter *in that frame* in the same sense that a pencil has a shorter shadow when you turn it, and the reason both observers can consistently say the other's rod is short is that they are disagreeing about which events are simultaneous. The transformation describes **the structure of space and time**. It is not a theory of rods and clocks; it is a statement about the arena they sit in, and it therefore applies to *everything* — nuclear forces, particle lifetimes, chemical clocks, things nobody had thought of — and not merely to electromagnetically bound matter.

That difference is not philosophical decoration. It is what makes the theory *predictive* outside its domain of construction. Lorentz's contraction is a claim about electromagnetically bound rods, and there is no reason it should apply to a decaying muon; Einstein's is a claim about the coordinates themselves, and it applies to the muon whether or not anything holding it together is electromagnetic. Muons in the atmosphere do live longer by exactly γ , and the weak interaction that decays them was not discovered for another thirty years.

And it is why the ether then had nothing left to do. Lorentz's ether was *necessary* to his derivation: without a medium, there is nothing for a rod to move through and no mechanism to contract it. Einstein's derivation never mentions it, so the ether becomes an object with no role in any calculation, detectable by no experiment, and postulated for no reason. It did not have to be refuted. It had to be noticed to be idle.

▼ Grind box — how close was Lorentz, exactly?

Uncomfortably close, and the gap is instructive. Take Lorentz's own 1904 apparatus and ask what it gets right.

Right: the algebra. (2.1.42) is character for character what Chapter 2.2 will derive. The contraction factor, the local time, the group property, the invariance of Maxwell's equations under the substitution — all correct, all published before 1905.

Right: the experimental predictions, for everything then measured. Michelson–Morley, Fizeau, aberration, the first-order optical experiments — Lorentz's theory reproduces them all.

Wrong: the status of t' . Because it is a substitution rather than a time, Lorentz had no account of time dilation as a physical phenomenon, and no reason to expect a moving clock of *any* construction to run slow. Ask him about an unstable particle and the theory is silent.

Wrong: the asymmetry. In Lorentz's scheme the contraction is real for a rod moving through the ether and not for a rod at rest in it, so the two frames are not equivalent even though no experiment can tell them apart. Einstein's symmetry between the frames is exact and is the reason each observer finds the other's rods short — a statement Lorentz's framework cannot make, since one of the two rods is really contracted and the other really is not.

Wrong: the generality. Lorentz's contraction was derived from a hypothesis about electromagnetic forces inside matter. Nothing in it covers gravitational binding, nuclear binding, or the decay rate of a particle. Every one of those obeys the same γ anyway, which on Lorentz's account would be an extraordinary coincidence and on Einstein's is a tautology.

The lesson generalises past this episode and is worth carrying: **having the right equations is not the same as having the right theory.** Two accounts can agree on every equation and every measured number and still differ in what they say the symbols mean — and that difference shows up the moment you try to apply the theory somewhere it was not built for. Lorentz's electrodynamics and Einstein's relativity were empirically indistinguishable in 1905. They stopped being indistinguishable the moment anyone measured a particle lifetime.

7 · The two postulates

Everything above is the case for the prosecution. Here is what Einstein proposed instead, stated in the form Chapter 2.2 will use as input.

THE POSTULATES OF SPECIAL RELATIVITY

Postulate 1 (the principle of relativity). *The laws of physics take the same form in all inertial frames.* No experiment performed within a uniformly moving laboratory can reveal that laboratory's velocity, and this applies to **all** the laws — mechanical, electromagnetic, and any not yet discovered — not merely the mechanical ones.

Postulate 2 (the invariance of c). *Light propagates in vacuum with a definite speed c , the same in all inertial frames, independent of the state of motion of the emitting body.*

Notice how little that is. There is no ether, no statement about what light is, no model of matter, no dynamics, and no mechanism. Two sentences, one of which was universally accepted already. Chapter 2.2 shows that the pair of them determines the transformation between inertial frames essentially uniquely — and everything in Part II after that is consequences.

7.1 · Postulate 1 is not new

It is §1.4's principle, with one word changed: *all* the laws, not just the mechanical ones. That single word is what rules out branch (A) of §4.3, where mechanics obeyed the principle and electromagnetism did not. Given (2.1.18) — Maxwell's equations demonstrably changing form under (2.1.1) — asserting Postulate 1 forces you into branch (C): the Galilean transformation must go.

7.2 · Postulate 2 is the radical one, and it is not a definition

This is where the strangeness lives, and it is worth being precise about why.

Consider a lamp on the front of a train moving at v and a lamp on the platform, flashing as they pass. Postulate 2 says both flashes travel at c relative to the platform *and* at c relative to the train. Not $c + v$ and c . Not c and $c - v$. The same c , measured by both observers, for both flashes. (2.1.2) — velocities add — is being denied, for light, flatly.

Two objections need answering.

"Isn't this just a convention about how to set clocks?" No. There is a real convention lurking nearby — synchronising distant clocks requires a rule, and any rule involves an assumption about one-way light speed — and Chapter 2.3 will treat it carefully. But Postulate 2 has empirical content independent of any such convention, because the *round-trip* speed is measurable without synchronising anything: one clock, one mirror, one pulse. The claim that the two-way speed is the same in all frames and independent of the source's motion is a fact about the world, and it can be checked.

"Isn't the source-independence part obvious for a wave?" For a wave in a medium, yes — sound from a moving whistle travels at 343 m s^{-1} through the air regardless of the whistle, since once the disturbance is launched the medium takes over and forgets the source. That is precisely why the ether was attractive: it delivers source-independence for free. Postulate 2 asserts source-independence *with no medium to enforce it*, which is a much stranger claim, and it also asserts frame-independence, which no medium theory delivers at all.

So Postulate 2 is an empirical claim and must be tested. It has been, exhaustively. Worked example 2 gives the sharpest version: photons emitted by particles moving at $0.99975c$ travel at c , not at $1.99975c$, to within a part in 10^4 .

HANDING OFF TO CHAPTER 2.2

Two postulates. One transformation to find. The question Chapter 2.2 asks is exactly this: *what transformation between inertial coordinates could possibly satisfy both?* — and the answer, remarkably, is that there is essentially only one. Linearity is forced by the requirement that free particles move in straight lines in every frame; the constant in it is forced by demanding a speed that comes out the same in all frames; and what drops out is (2.1.42), with $t' \neq t$ and everything that follows from it.

You already know one thing about the answer that Chapter 2.2 will have to reproduce. Whatever the transformation is, it must produce a contraction by $\sqrt{1 - \beta^2}$ along the direction of motion and nothing across it, because that is precisely the combination (2.1.40) needs to explain Michelson–Morley. The experiment has already told you the shape of the answer. What it could not tell you is why.

8 · Worked examples

WORKED EXAMPLE 1 — THE 1887 APPARATUS, ALL THE WAY THROUGH

Take the Michelson–Morley interferometer as built: effective arm length $L = 11.0$ m in each arm, sodium light at $\lambda = 590$ nm, and the ether at rest relative to the Sun so that the wind speed is the Earth's orbital speed $v = 30.0$ km s⁻¹. Compute both round-trip times, their difference, and the fringe shift on a 90° rotation. Compare with the experimental bound.

Step 1 — the small parameter. With $c = 2.99792 \times 10^8$ m s⁻¹,

$$\beta = \frac{3.00 \times 10^4}{2.99792 \times 10^8} = 1.0007 \times 10^{-4}, \quad \beta^2 = 1.0014 \times 10^{-8}.$$

Already the shape of the problem is clear: we are chasing a part in 10^8 .

Step 2 — the two round-trip times. The no-wind time is $2L/c = 22.0/2.99792458 \times 10^8 = 7.33841009 \times 10^{-8}$ s = 73.38410094 ns. Then from (2.1.23) and (2.1.25),

$$t_{\parallel} = \frac{73.38410094 \text{ ns}}{1 - 1.0014 \times 10^{-8}} = 73.384101678 \text{ ns},$$

$$t_{\perp} = \frac{73.38410094 \text{ ns}}{\sqrt{1 - 1.0014 \times 10^{-8}}} = 73.384101311 \text{ ns}.$$

They agree to nine significant figures. This is why nobody attempted to time the two beams separately.

Step 3 — the difference. Subtracting nine matching digits is a bad idea numerically, so use the exact rearrangement

$$\Delta t = \frac{2L}{c} \cdot \frac{\beta^2}{(1 - \beta^2)(1 + \sqrt{1 - \beta^2})} = 3.6743 \times 10^{-16} \text{ s},$$

which is 367 attoseconds. Cross-check against the leading-order estimate (2.1.30): $L\beta^2/c = 11 \times 1.0014 \times 10^{-8}/2.99792 \times 10^8 = 3.6743 \times 10^{-16}$ s. Agreement to five figures. ✓

Step 4 — turn it into a distance. A time difference is invisible; a path difference is not. Multiply by c :

$$c \Delta t = 1.1015 \times 10^{-7} \text{ m} = 110.15 \text{ nm} = 0.1867 \lambda.$$

The two beams arrive about a fifth of a wavelength out of step. This is the entire physical content of the experiment: **a fifth of a wavelength, from a 367-attosecond delay, produced by the Earth moving at a ten-thousandth of the speed of light.**

Step 5 — the fringe shift on rotation. Rotating exchanges the arms, so the path difference swings from $+0.1867\lambda$ to -0.1867λ and the fringes move by twice that:

$$\Delta N = \frac{2c \Delta t}{\lambda} = 2 \times 0.1867 = 0.373 \text{ fringes}.$$

Or directly from (2.1.34): $\Delta N \approx 2L\beta^2/\lambda = 2(11)(1.0014 \times 10^{-8})/(5.90 \times 10^{-7}) = 0.373$. ✓ (The exact expression gives 0.373398, the expansion 0.373398; they differ in the ninth decimal.)

Step 6 — compare. ▀ Observed: less than 0.01 fringes. Predicted: 0.373. The ratio is

$$\frac{0.373}{0.01} = 37.3.$$

To hide the prediction under the bound you would need β^2 smaller by a factor of 37, i.e. v smaller by $\sqrt{37} = 6.1$, i.e. an ether wind below 4.9 km s^{-1} — a sixth of the Earth's orbital speed, and a speed the Earth cannot have relative to a fixed frame at all times of year.

A number worth having. The Solar System moves at about 220 km s^{-1} around the galactic centre, and at about 370 km s^{-1} relative to the frame in which the cosmic microwave background is isotropic. Had the ether been at rest in either, the 1887 apparatus would have shown 20 fringes or 57 fringes respectively — the pattern would have swept past the crosshairs dozens of times as the table turned. There is no plausible ether frame for which this experiment is a close call.

WORKED EXAMPLE 2 — WHAT GALILEAN ADDITION DEMANDS OF LIGHT, AND WHAT IS MEASURED

Suppose light obeys (2.1.2) like everything else. Work out what a laboratory should measure for light from a moving source, and compare with experiment.

Step 1 — the prediction. If light leaves its source at c relative to the source, and velocities add by (2.1.2), then a lab in which the source moves at v directly towards the detector measures

$$c_{\text{lab}} = c + v,$$

and $c - v$ for a receding source. This is the "emission theory" or "ballistic" hypothesis, and it is what you get if you take the Galilean transformation seriously and drop the ether. Note that it is *not* the ether theory — the ether predicts c relative to the ether regardless of the source. The two hypotheses are different and both are testable.

Step 2 — the astronomical test. De Sitter's argument, 1913, and it costs nothing but arithmetic. Take a spectroscopic binary star at distance D whose components orbit at speed v_{orb} . Light emitted while a star approaches us travels (on this hypothesis) at $c + v$; half an orbit later, receding, at $c - v$. The difference in arrival time after a journey D is

$$\Delta t_{\text{arr}} = \frac{D}{c - v} - \frac{D}{c + v} = \frac{2Dv}{c^2 - v^2} \approx \frac{2Dv}{c^2}.$$

Put in numbers for a typical system — $D = 100$ light years $= 9.46 \times 10^{17}$ m, $v_{\text{orb}} = 100$ km s $^{-1}$:

$$\Delta t_{\text{arr}} = \frac{2Dv}{c^2} = \frac{2 \times 9.46 \times 10^{17} \times 10^5}{8.99 \times 10^{16}} = 2.11 \times 10^6 \text{ s} = 24 \text{ days}.$$

Compare that with an orbital period of, say, 5 days. The scrambling is *five times the period*: fast-phase light would overtake slow-phase light, the star would appear at several places in its orbit at once, and the spectroscopic velocity curve would be unrecognisable — not merely distorted but multivalued. Nothing of the kind is seen. Writing the hypothesis as $c_{\text{lab}} = c + kv$ multiplies the estimate by k , and requiring the distortion to stay within the per-cent level at which binary orbits are observed to fit Kepler's laws needs $k \lesssim 0.01 \times (5/24) \approx 2 \times 10^{-3}$. ■ De Sitter's published bound was of this order; ■ Brecher's 1977 analysis of X-ray binaries, where the timing is far sharper, tightened it to 2×10^{-9} .

Step 3 — the laboratory test, and it is brutal. ■ At CERN in 1964, Alväger and collaborators produced neutral pions in a proton beam and used the photons from $\pi^0 \rightarrow \gamma\gamma$. The pions were moving at $\beta_\pi = 0.99975$ — as close to a light-speed source as anything ever built — and the photons' speed was measured by time of flight over a known baseline.

The Galilean prediction for photons emitted forwards is unambiguous:

$$c_{\text{lab}} = c + v_\pi = (1 + 0.99975)c = 1.99975c = 5.995 \times 10^8 \text{ m s}^{-1}.$$

Very nearly *double* the speed of light. What was measured was

$$c_{\text{lab}} = (2.9977 \pm 0.0004) \times 10^8 \text{ m s}^{-1},$$

i.e. c to within 1.3 parts in 10^4 . Writing the hypothesis as $c_{\text{lab}} = c + k v_{\pi}$, the measurement gives $k = (-0.7 \pm 1.3) \times 10^{-4}$: consistent with zero, and excluding $k = 1$ by about **seven thousand standard deviations**.

What to take from this. Both experiments are quoted results, not derivations — the derivation in each case is the prediction they are compared against, which is the arithmetic above. And note the logical structure of the pair. The ether hypothesis is killed by Michelson–Morley plus aberration plus Fizeau; the emission hypothesis, which survives all three, is killed by de Sitter and by π^0 decay. Between them the two families of experiment eliminate every way of keeping Galilean velocity addition for light. Postulate 2 is what is left standing.

9 • Your turn

PROBLEM 1 — FITZGERALD'S FIX, AND WHAT IT CANNOT FIX

(a) Redo (2.1.33) assuming the arm along the wind contracts by $\sqrt{1 - \beta^2}$ while the transverse arm is unchanged, keeping $L_1 \neq L_2$. Show the fringe shift on rotation vanishes identically, for any β and any arm lengths. (b) The cancellation in (2.1.40) is exact, so there is no residual at order β^4 to find in the equal-arm case. Where, then, does an observable β -dependence survive? Compute the total time difference between the two beams for *unequal* arms with contraction, and show it depends on β . (c) Estimate the size of the effect for $L_1 - L_2 = 16$ cm, $\lambda = 590$ nm, as the Earth's speed relative to a hypothetical ether changes from 30 to 60 km s⁻¹ over six months. (d) What extra ingredient removes even this?

– Solution

(a) With contraction, the arm currently along the wind has ether-frame length $L_i \sqrt{1 - \beta^2}$, so by (2.1.40) its round-trip time is $(2L_i/c)/\sqrt{1 - \beta^2}$ — the same functional form as the transverse arm. Hence

$$\Delta t_{\text{before}} = \frac{2L_1/c}{\sqrt{1 - \beta^2}} - \frac{2L_2/c}{\sqrt{1 - \beta^2}} = \frac{2(L_1 - L_2)/c}{\sqrt{1 - \beta^2}},$$

and after rotation arm 2 is along the wind and arm 1 across it, giving *the same expression* — because both arms now have identical round-trip formulas and only the labels have swapped. The difference is zero identically. The fringe shift on rotation is zero for any β , any L_1 , any L_2 . Michelson–Morley is fully explained, not merely to leading order.

(b) Look at what survives: the beams still differ in arrival time, by

$$\Delta t = \frac{2(L_1 - L_2)}{c} \cdot \frac{1}{\sqrt{1 - \beta^2}} = \frac{2(L_1 - L_2)}{c} \gamma \approx \frac{2(L_1 - L_2)}{c} \left(1 + \frac{1}{2}\beta^2\right).$$

This is not zero and it *depends on* β . Rotating the apparatus does not change it, so Michelson–Morley cannot see it. But the Earth's speed relative to any putative ether changes over the year as its orbital velocity swings around, so the fringe pattern should slowly drift with the seasons even with the apparatus never moved. That is precisely the experiment Kennedy and Thorndike performed in 1932, with deliberately unequal arms, and it is the natural complement to Michelson–Morley: MM varies the *direction* at fixed speed, KT varies the *speed* at fixed direction.

(c) The fringe count is $N = c\Delta t/\lambda = 2(L_1 - L_2)\gamma/\lambda$, so the change between two speeds is

$$\Delta N = \frac{2(L_1 - L_2)}{\lambda} (\gamma_2 - \gamma_1) \approx \frac{(L_1 - L_2)}{\lambda} (\beta_2^2 - \beta_1^2).$$

With $\beta_1 = 1.0007 \times 10^{-4}$ and $\beta_2 = 2.0014 \times 10^{-4}$, $\beta_2^2 - \beta_1^2 = 3.004 \times 10^{-8}$, and

$$\Delta N = \frac{0.16 \times 3.004 \times 10^{-8}}{5.90 \times 10^{-7}} = 8.1 \times 10^{-3} \text{ fringes.}$$

Small — under a hundredth of a fringe — but a stable interferometer watched over months can reach it, and ■ Kennedy and Thorndike saw nothing.

(d) Time dilation. The fringe count is a path difference divided by a wavelength, and the wavelength is set by the *source*, which is moving too. If a moving clock runs slow by γ , then the light emitted by a moving atom has its period stretched by γ and its wavelength stretched by γ as well, so $\lambda \rightarrow \gamma\lambda_0$ and

$$N = \frac{c \Delta t}{\lambda} = \frac{2(L_1 - L_2)\gamma}{\gamma\lambda_0} = \frac{2(L_1 - L_2)}{\lambda_0},$$

independent of β . Contraction alone explains Michelson–Morley; contraction *and* time dilation together are needed for Kennedy–Thorndike. Historically this is the cleanest demonstration that length contraction is not a complete story: the two experiments together force both effects, and both drop out of Chapter 2.2 from the two postulates without either being assumed. Add the observed isotropy of c from Michelson–Morley and you have, between the three experiments, enough to pin the Lorentz transformation down uniquely — which is why this trio is sometimes called the experimental basis of special relativity.

PROBLEM 2 — THE TRANSFORMED WAVE EQUATION, AND WHICH TERM IS THE CULPRIT

(a) Repeat §4 for a Galilean boost of speed v along x applied to the full three-dimensional wave equation $\nabla^2\phi = c^{-2}\partial_t^2\phi$, and write the result. (b) Identify precisely which term breaks the symmetry, and give two independent arguments that it cannot appear in a law obeying the principle of relativity. (c) Show that if you keep only terms of order β^0 you recover the original equation — and say why that is not a rescue.

– Solution

(a) The boost affects only x and t , so $\partial_y = \partial_{y'}$ and $\partial_z = \partial_{z'}$ are untouched and (2.1.17) applies verbatim to the rest. Substituting into $\nabla^2\phi = c^{-2}\partial_t^2\phi$:

$$\frac{\partial^2\phi}{\partial x'^2} + \frac{\partial^2\phi}{\partial y'^2} + \frac{\partial^2\phi}{\partial z'^2} = \frac{1}{c^2} \left[\frac{\partial^2\phi}{\partial t'^2} - 2v \frac{\partial^2\phi}{\partial x' \partial t'} + v^2 \frac{\partial^2\phi}{\partial x'^2} \right],$$

i.e.

$$(1 - \beta^2) \frac{\partial^2\phi}{\partial x'^2} + \frac{\partial^2\phi}{\partial y'^2} + \frac{\partial^2\phi}{\partial z'^2} + \frac{2v}{c^2} \frac{\partial^2\phi}{\partial x' \partial t'} - \frac{1}{c^2} \frac{\partial^2\phi}{\partial t'^2} = 0.$$

Two symptoms, both visible: the x' derivative now has a different coefficient from the y' and z' ones, so **space is no longer isotropic**; and there is a cross term.

(b) The culprit is $\frac{2v}{c^2} \frac{\partial^2\phi}{\partial x' \partial t'}$. Two arguments.

Parity. Under $x' \rightarrow -x'$ every other term is unchanged (each carries an even number of x' derivatives) while the cross term changes sign, since it carries exactly one. So the equation distinguishes $+x'$ from $-x'$ — it knows which way the wind blows. A law valid in a frame with no preferred direction cannot contain such a term.

Boost-parameter dependence. The coefficient is proportional to v , a quantity that describes the relationship between two frames rather than anything in the physics. Any law whose coefficients depend on the observer's velocity is by definition not the same law for all observers, which is the negation of Postulate 1.

Note that the modified coefficient $1 - \beta^2$ is *also* fatal but is a weaker symptom: it is even in v , so it survives parity, and one could imagine absorbing it by rescaling x' . You cannot absorb the cross term that way — it is the one that cannot be rescaled away, and eliminating it is exactly what forces t' to depend on x in Chapter 2.2. That is worth noticing now: the term that breaks the symmetry is the term that mixes space and time derivatives, and the repair will be a transformation that mixes space and time coordinates.

(c) Setting $\beta = 0$ in the boxed equation returns $\nabla^2\phi = c^{-2}\partial_t^2\phi$ exactly, and to first order in β only the cross term survives, at relative size $2v/c \cdot (\omega/k)/c \sim 2\beta$. So for $v \ll c$ the equation is *nearly* unchanged,

which is precisely why the discrepancy went unnoticed for forty years — every terrestrial source moves at $\beta \lesssim 10^{-4}$.

But it is not a rescue, for two reasons. First, "approximately invariant" is not a coherent notion for a symmetry principle: either the laws are the same in all frames or they are not, and if they are not there exists in principle a measurement that identifies your frame, no matter how hard it is. Second, and decisively, the experiments of §5 and Worked example 2 are *designed* to reach the order at which the discrepancy lives — Michelson–Morley to β^2 , the π^0 experiment to $\beta \approx 1$ where the "small" correction is a factor of two. At those sensitivities the effect is not small, and it is absent.

PROBLEM 3 — STELLAR ABERRATION AND THE DRAG HYPOTHESIS

(a) Derive the aberration angle for a star directly overhead from scratch, using the moving telescope picture, and evaluate it for the Earth's orbital speed. (b) Explain quantitatively why an observer sees the star's apparent position trace out a small ellipse over a year, and what the ellipse's shape depends on. (c) Explain precisely why complete ether drag is incompatible with the observation, and why *partial* drag does not obviously save it either. (d) A rain analogy is often used. State the analogy, then say exactly where it fails once Postulate 2 is adopted.

– Solution

(a) Let the telescope tube have length ℓ and the Earth move sideways at v . Light enters the objective and takes ℓ/c to reach the eyepiece; in that time the eyepiece has moved $v\ell/c$ in the direction of motion. For the light to land on the eyepiece, the tube must be tilted forwards by an angle θ with

$$\tan \theta = \frac{v\ell/c}{\ell} = \frac{v}{c}.$$

The tube length cancels, which is the sign of a real effect rather than an instrumental one. With $v = 29.78 \text{ km s}^{-1}$, $\theta = 9.9335 \times 10^{-5} \text{ rad}$; multiplying by $206265'' \text{ rad}^{-1}$ gives $\theta = 20.49''$. ▀ The measured constant of aberration is $20.4955''$.

(b) The tilt is always *towards the instantaneous direction of the Earth's motion*, and that direction rotates through 360° in a year as the Earth goes round its orbit. So the apparent position of any star swings around a closed curve of angular radius v/c . A star at the pole of the ecliptic sees the Earth's velocity vector sweep out a full circle in the plane perpendicular to the line of sight, and its aberration ellipse is a *circle* of radius $20.5''$. A star in the plane of the ecliptic sees only the component of the Earth's velocity perpendicular to the line of sight, which oscillates back and forth, and its ellipse degenerates to a *line* of half-length $20.5''$. In between, the semi-minor axis is $20.5'' \sin \lambda$ with λ the star's ecliptic latitude. That pattern — same semi-major axis for every star; semi-minor axis depending only on ecliptic latitude and not at all on distance — is the signature that identifies aberration and distinguishes it from parallax, whose size depends on distance and whose phase is shifted by a quarter year.

(c) The derivation in (a) assumed the light travels in a straight line at c in a frame in which the telescope is *moving*. If the ether were completely dragged along by the Earth, then within the dragged region light would propagate in the telescope's own rest frame, the eyepiece would not move relative to the medium during the transit, and there would be no tilt. You would see no annual aberration at all. It is observed, at full strength, so complete drag is excluded.

Partial drag does not rescue it either, and for an instructive reason. If the drag is partial, the aberration should depend on how much dragged medium the light passes through — so it should differ for observations made through a long column of air versus a short one, or better, it should change if you fill the telescope tube with water, since water has a Fresnel drag coefficient of 0.44 by (2.1.39) and would drag the light substantially. ▀ Airy performed exactly this experiment in 1871 with a water-filled telescope and found the aberration *unchanged* — the same $20.5''$. A drag model has to explain why filling the instrument with a strongly dragging medium changes nothing, and the honest answer within ether

theory is a conspiracy of cancellations. (Relativity gets Airy's null result immediately: aberration is a property of the transformation between the source's frame and the observer's, and has nothing to do with what the light passes through on the way in.)

(d) The analogy. Running through vertically falling rain, you must tilt your umbrella forwards; the faster you run, the more you tilt. The tilt angle satisfies $\tan \theta = v_{\text{you}}/v_{\text{rain}}$, which is (2.1.37) with the rain speed in place of c .

Where it fails. The rain calculation is a Galilean velocity addition: in your frame the raindrops have velocity $(-v_{\text{you}}, -v_{\text{rain}})$, and the tilt is the direction of that resultant. It also predicts that the drops arrive *faster* in your frame, $\sqrt{v_{\text{you}}^2 + v_{\text{rain}}^2}$. For light, Postulate 2 forbids that: the photons arrive at c , not at $\sqrt{v^2 + c^2}$. So the correct relativistic aberration formula cannot be the vector-addition one, and it is not — Chapter 2.2 derives $\cos \theta' = (\cos \theta - \beta)/(1 - \beta \cos \theta)$, which agrees with (2.1.37) to first order in β and differs at second order. Since $\beta \approx 10^{-4}$ for the Earth, the correction enters at relative order $\beta^2 = 10^{-8}$ and reaches at most $\beta^2/4 = 2.5 \times 10^{-9}$ rad, i.e. 5×10^{-4} arcseconds — far below the precision of nineteenth-century astrometry, which is why the naive derivation gave the right answer for two centuries and why aberration could not have settled the question by itself. It is measurable now, and the relativistic formula is the one that matches.

PROBLEM 4 — HOW FAST WOULD YOU HAVE TO GO?

(a) Find the wind speed v that would produce a fringe shift of exactly 1.000 in the 1887 apparatus ($L = 11$ m, $\lambda = 590$ nm). Do it first from the leading-order formula, then check against the exact expression. (b) Comment on whether that speed is achievable or plausible. (c) Suppose instead you fix $v = 30$ km s⁻¹ and ask how long the arms would have to be for a one-fringe shift. Is *that* achievable? (d) What does the comparison tell you about why the experiment was done in 1887 and not in 1830?

– Solution

(a) From (2.1.34) at leading order, $\Delta N \approx 2L\beta^2/\lambda$, so $\Delta N = 1$ requires

$$\beta = \sqrt{\frac{\lambda}{2L}} = \sqrt{\frac{5.90 \times 10^{-7}}{22.0}} = \sqrt{2.682 \times 10^{-8}} = 1.6377 \times 10^{-4},$$

$$v = \beta c = 1.6377 \times 10^{-4} \times 2.99792 \times 10^8 = 4.91 \times 10^4 \text{ m s}^{-1} = 49.1 \text{ km s}^{-1}.$$

Solving the exact expression (2.1.34) numerically gives $\beta = 1.637626 \times 10^{-4}$, $v = 49.0948$ km s⁻¹ — the same to six figures, since the correction is $O(\beta^2) = 10^{-8}$.

(b) 49 km s⁻¹ is only 1.6 times the Earth's orbital speed. It is laughably out of reach for a laboratory — no apparatus has ever been moved at anything close — but it is *utterly ordinary* as an astronomical speed. That is the whole point of the experiment's design: it does not need you to move the apparatus, because the Earth is already moving. And it means the null result is not marginal. For the ether to hide, its rest frame must lie within about 5 km s⁻¹ of the Earth at all times of year — impossible, since the Earth's own velocity changes by 60 km s⁻¹ between January and July, so any fixed frame is at least 30 km s⁻¹ away from the Earth for half the year.

(c) Rearranging for L at $\Delta N = 1$ with $\beta = 1.0007 \times 10^{-4}$:

$$L = \frac{\lambda}{2\beta^2} = \frac{5.90 \times 10^{-7}}{2 \times 1.0014 \times 10^{-8}} = 29.5 \text{ m}.$$

Under thirty metres of optical path, in each arm, held rigid to a fraction of a wavelength and rotatable. Michelson achieved 11 m by folding the beam back and forth eight times across a 1.5 m stone; tripling that is difficult but not absurd, and later workers did reach tens of metres. So the two routes to sensitivity are not symmetric: **you cannot change v at all, and you can change L by a factor of a few.** The experiment lives or dies on L , which is why every improvement in this line of work has been an improvement in effective path length — culminating in the modern optical resonators, where the light bounces 10^5 times and the effective path is kilometres.

(d) Because $\Delta N \approx 2L\beta^2/\lambda$ has $\beta^2 = 10^{-8}$ in it, and an experiment sensitive to one part in 10^8 was not buildable earlier. Everything that mattered had to arrive first: sufficiently monochromatic sources, high-quality half-silvered mirrors, mechanical isolation (the mercury float), and above all the realisation that interferometry converts a hopeless timing measurement into a feasible displacement measurement.

Michelson invented the instrument for this purpose. It is a general pattern worth noticing — *the second-order effect is where the interesting physics is, and second-order effects only become visible when someone builds a null-comparison instrument*. The same story runs through Cavendish's torsion balance, Eötvös's test of the equivalence principle (Chapter 3.1), and LIGO, whose optical layout is a Michelson interferometer with 4 km arms looking for a displacement of 10^{-18} m. The instrument that failed to find the ether became, a century later, the instrument that found gravitational waves.

You have the crisis, in a form you can state in three sentences and defend line by line.

One. Galilean relativity is exact for Newtonian mechanics. The transformation is $x' = x - vt$, $t' = t$; velocities subtract; accelerations are unchanged; forces built from relative positions and velocities are unchanged; therefore $\mathbf{F} = m\mathbf{a}$ has literally the same form in every inertial frame. And you have the crucial separation: the *principle* of relativity (P) and the *transformation* (G) that implements it are different claims, fused together for three centuries because $t' = t$ did not look like an assumption.

Two. Maxwell's equations produce a speed. Curl of Faraday, commute the derivatives, substitute Ampère–Maxwell, expand with $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ and use $\nabla \cdot \mathbf{E} = 0$, and out comes the wave equation of Chapter 0.8 with $c = 1/\sqrt{\mu_0 \epsilon_0} = 2.998 \times 10^8 \text{ m s}^{-1}$ — from a capacitor and a pair of current-carrying wires. That is why light is an electromagnetic wave.

Three. The two are incompatible, and the proof is three lines of chain rule. Under a Galilean boost the wave equation acquires a cross term $\propto \partial^2 \phi / \partial x' \partial t'$ and a modified coefficient $1 - \beta^2$. It changes form. So either the principle of relativity fails for electromagnetism, or Maxwell is wrong, or the Galilean transformation is wrong — and the first option makes a prediction that Michelson and Morley went looking for and did not find, at a predicted 0.373 fringes against a bound of 0.01.

You also have the anatomy of the effect, which matters more than the history: the parallel arm is slowed by γ^2 and the transverse by γ , the difference is second order in β , and the whole observable lives in the gap between those two exponents. And you know that FitzGerald's ad hoc contraction by $\sqrt{1 - \beta^2}$ cancels the effect *exactly*, that Lorentz had the transformation equations before Einstein, and that what 1905 contributed was not the algebra but the recognition that these are facts about space and time rather than about matter moving through a medium — which is why the ether then had nothing left to do.

Where this gets spent.

- **The two postulates** — Chapter 2.2, immediately. They are the entire input; the Lorentz transformation is the unique output, and every strange consequence — the relativity of simultaneity, time dilation, length contraction — arrives as a corollary rather than an assumption.
- **The Pythagoras triangle of (2.1.26)** — Chapter 2.2's light clock, where the identical triangle gives time dilation and the identical $1/\sqrt{1 - \beta^2}$ is named γ .
- **$t' \neq t$, and $\partial t' / \partial x \neq 0$** — Chapter 2.3, where the failure of simultaneity becomes geometry: events acquire an invariant separation $\Delta s^2 = c^2 \Delta t^2 - \Delta x^2$, and the light cone is the set of events (2.1.10) can reach.
- **"Both sides transform the same way, so the equality survives"** — the criterion isolated in §1's grind box — Chapter 2.4, where it is promoted to the definition of a tensor equation and becomes the reason the rest of this book is written in indices.

- **The wave equation and the constant c** – **Chapter 2.6**, the resolution. Maxwell's equations turn out to be exactly Lorentz invariant, with no modification whatever — they were relativistic all along, forty years before anyone knew what that meant, and it was Newtonian mechanics that had to be changed. That chapter also pays Chapter 1.1's outstanding debt: the field momentum $\epsilon_0 \mathbf{E} \times \mathbf{B}$ that repairs the third law's failure.
- **The whole argument's shape** → Chapter 3.1, which runs it again. A principle believed on excellent grounds (relativity) meets a fact believed on excellent grounds (gravitational and inertial mass are equal), they turn out to be incompatible, and the resolution is again that a piece of assumed structure — this time the flatness of spacetime — has to go. And Chapter 5.1 runs it a third time for quantum mechanics and relativity, where the resolution is that particle number cannot be conserved and fields become unavoidable.

One sentence to carry: *the speed in a wave equation belongs to the frame in which the equation holds, and Maxwell's equations refused to name that frame.* Everything in Part II is the consequence of taking that refusal at face value.